Lab 07 Manual

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SRT411NAA: Digital Data Analytics

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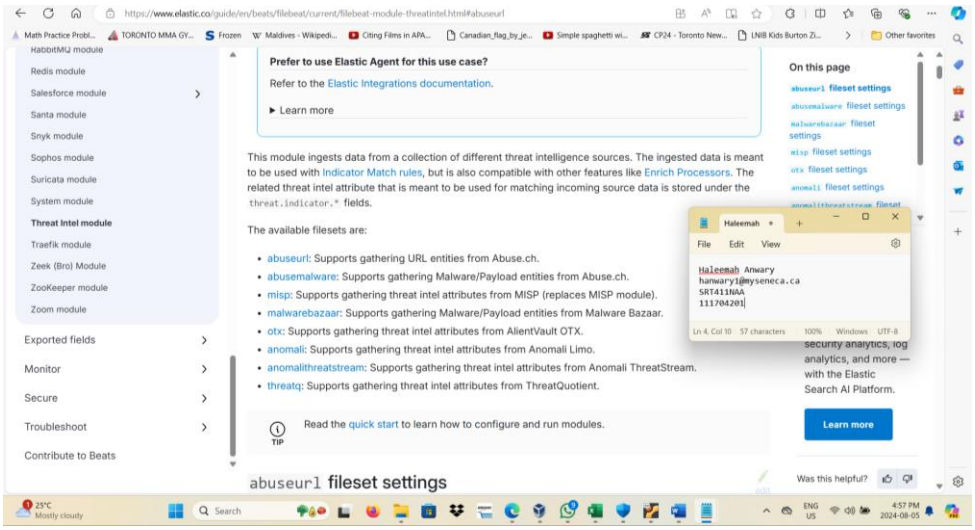
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Task 01 (Open-Source Threat Intelligence)

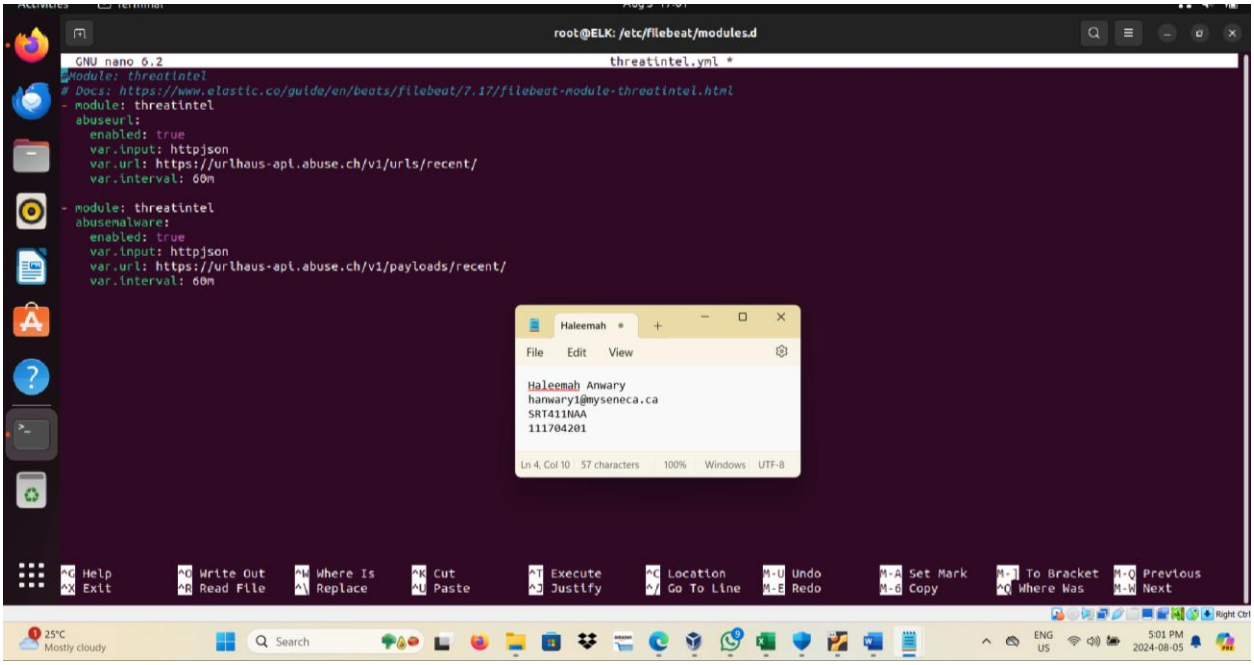
Here, we can find the collection of different threat intelligence sources provided by filebeat module.d and their respective configurations.



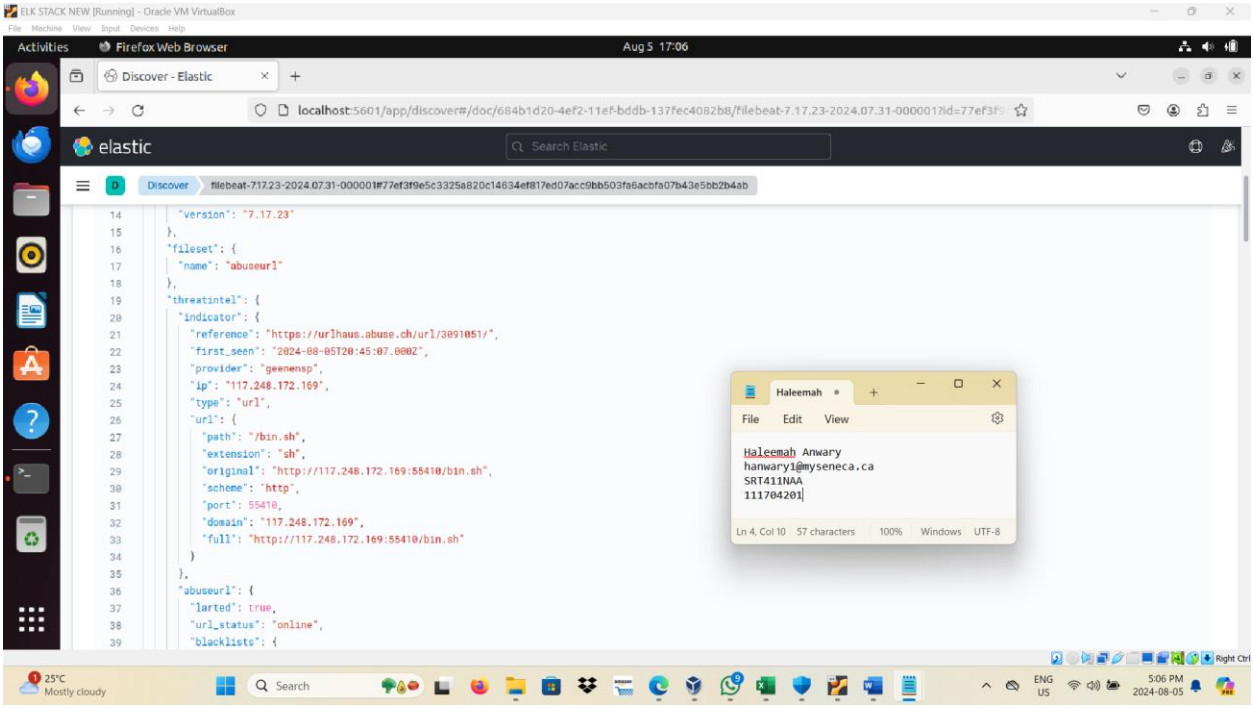
I enabled the threatintel module and configured It to extract abuseurl and abusemalware threat intelligence.

Abuseurl: this fileset contacts the abuse.ch API and fetches all new malicious URLs found the last 60 minutes.

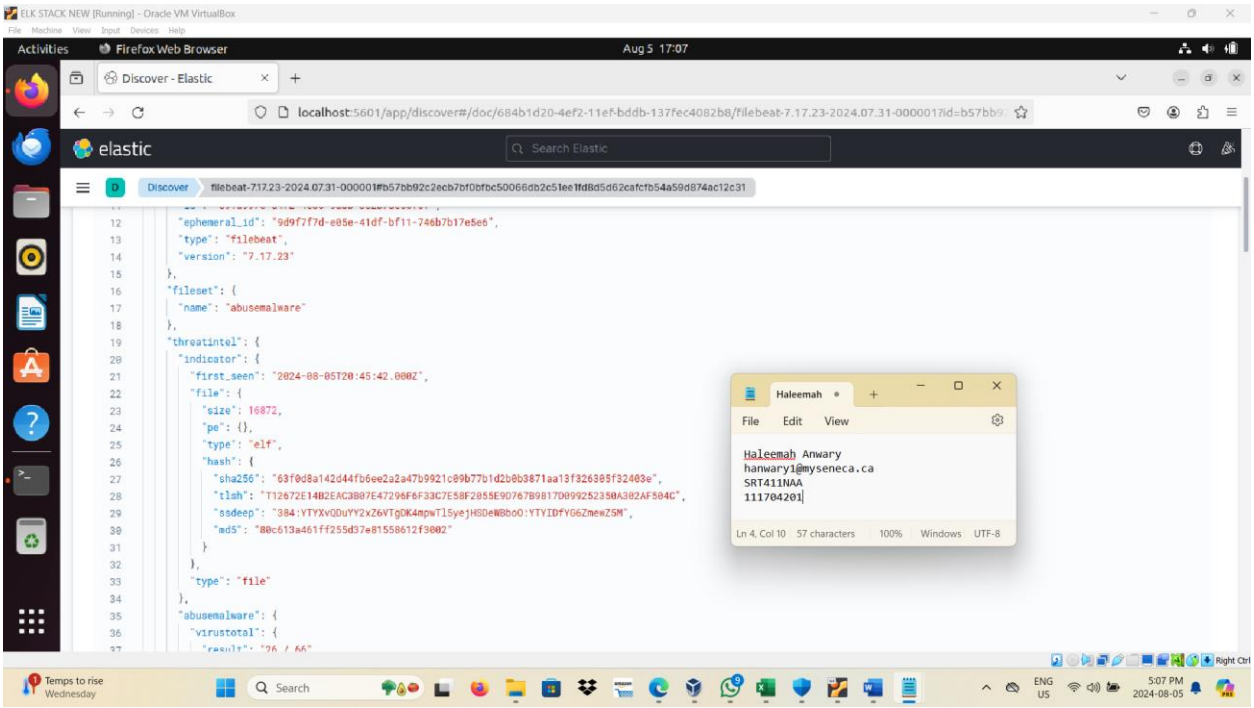
Abusemalware: this fileset contacts the Abuse.ch API and fetches all new malicious hashes found the last 60 minutes



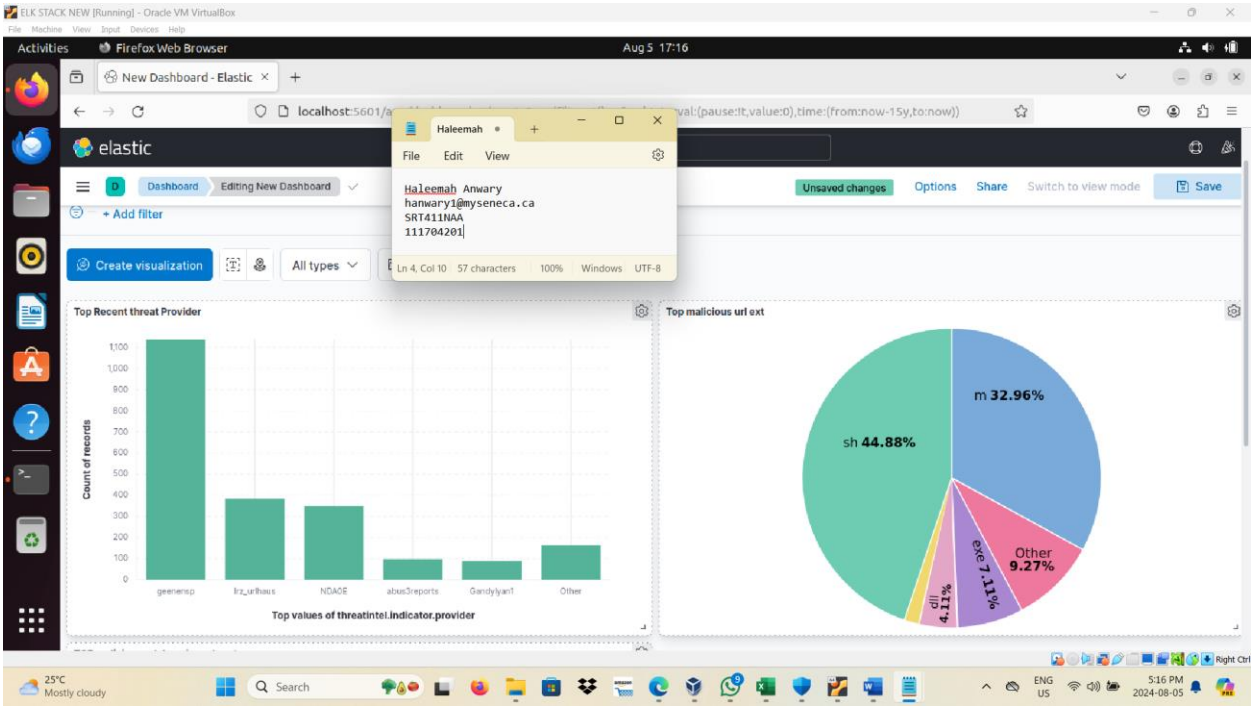
Abuseurl In Discover

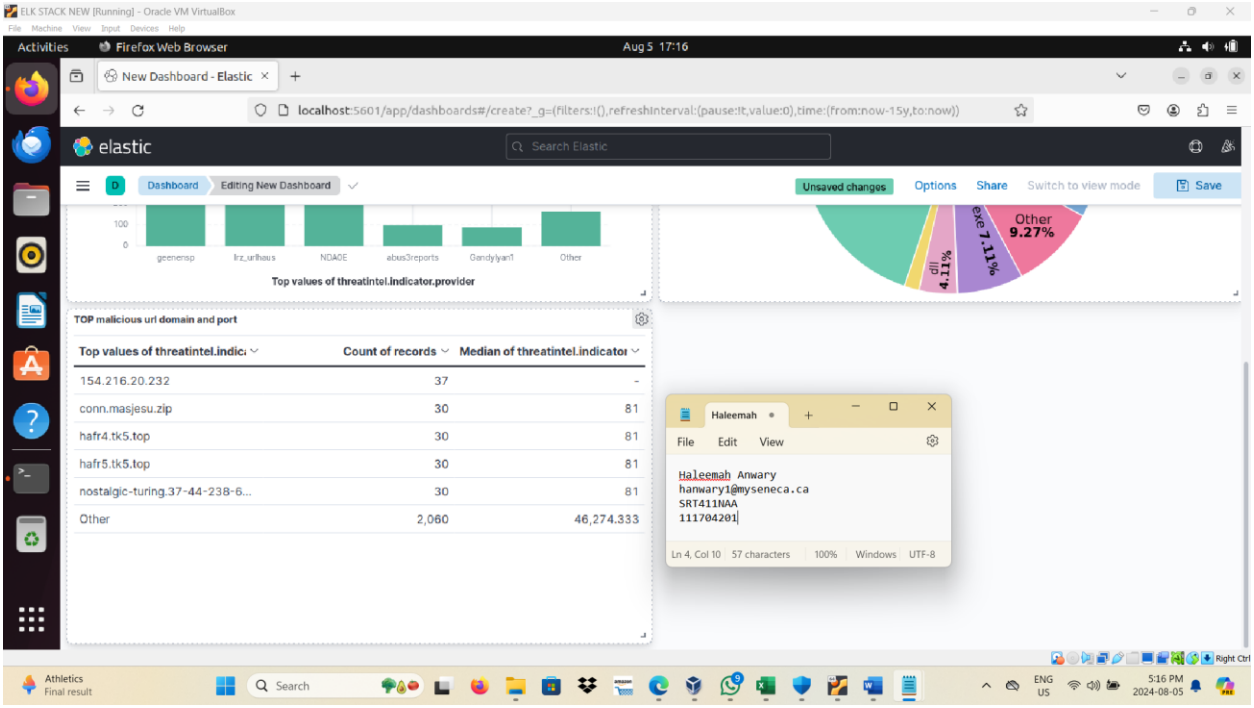


Abusemalware In Discover

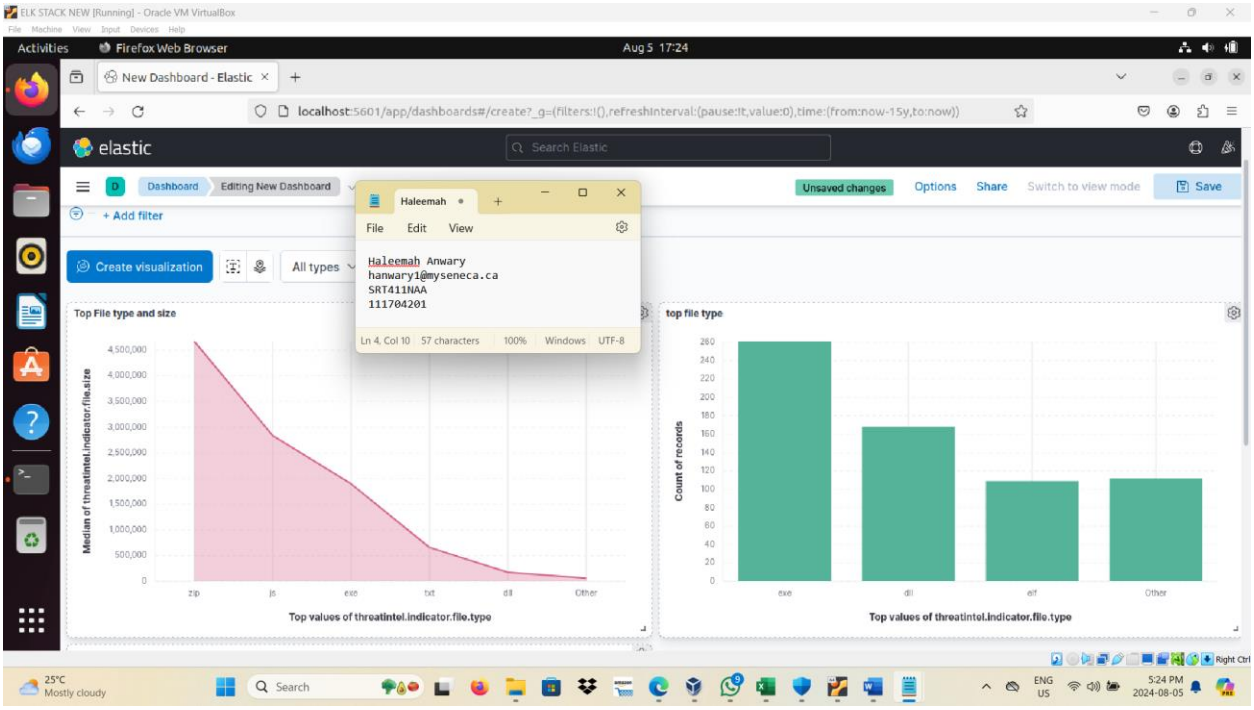


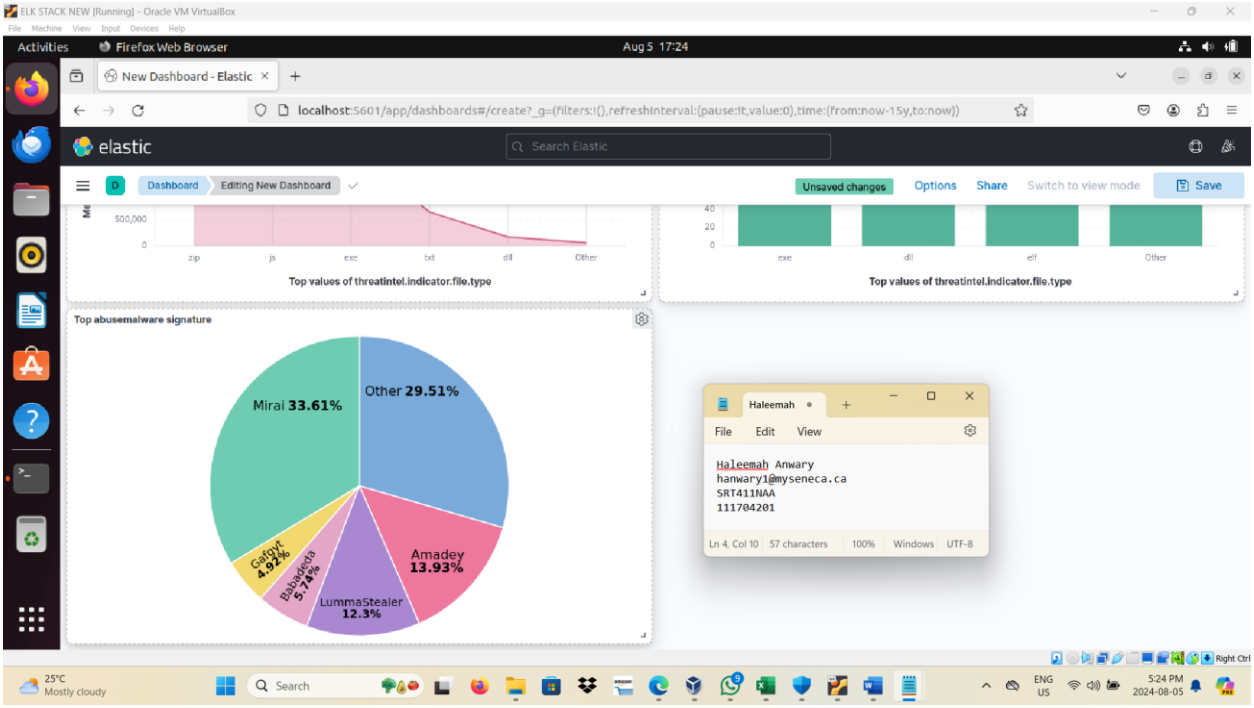
Abuseurl Dashboard





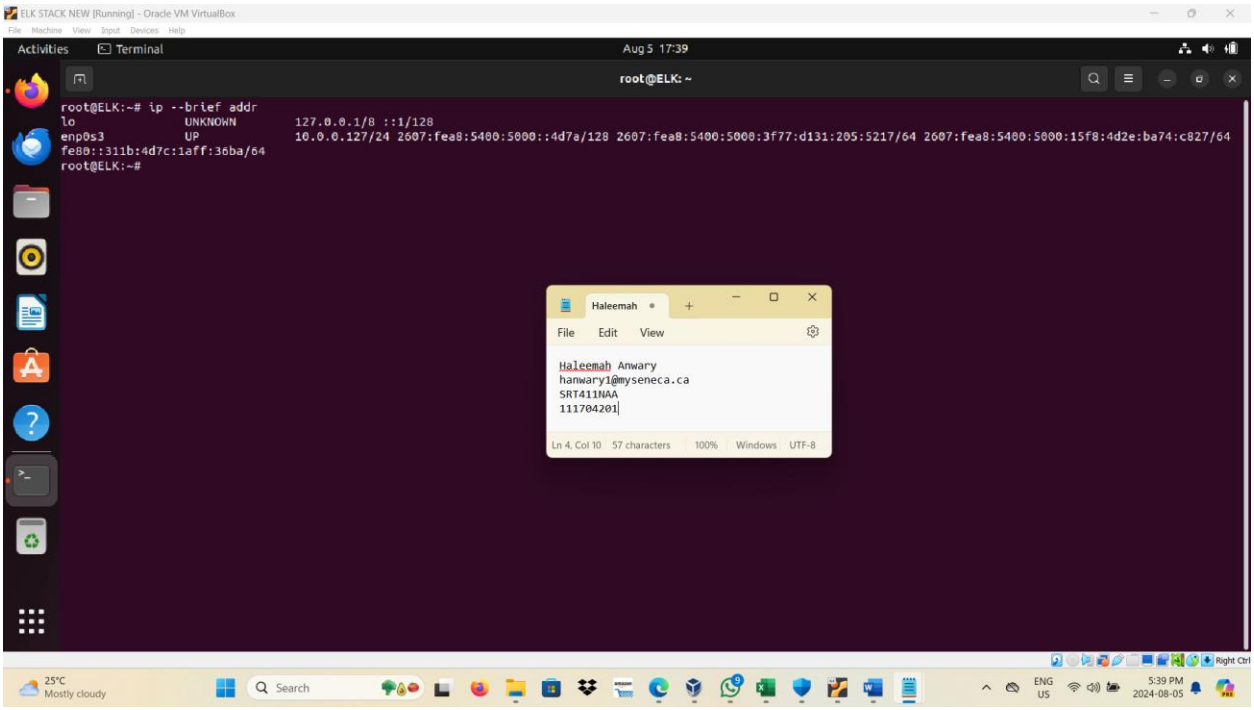
Abusemalware Dashboard



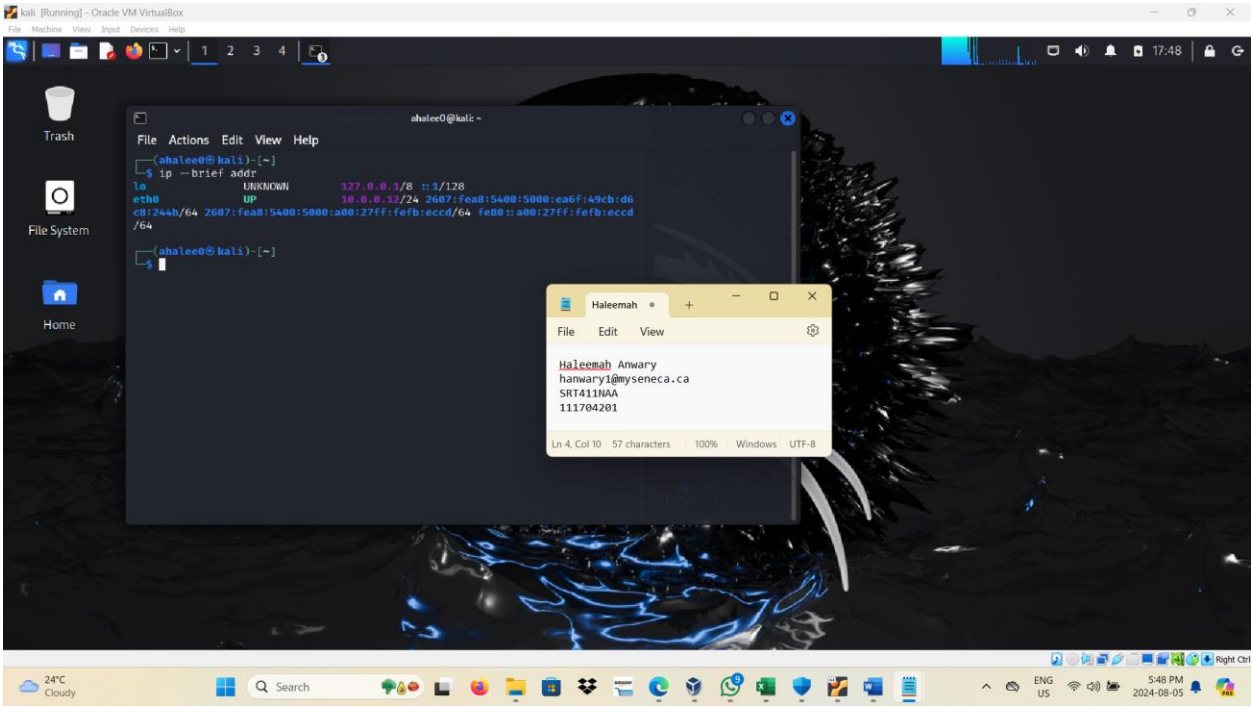


Task 02 (AutoInject Known Threats)

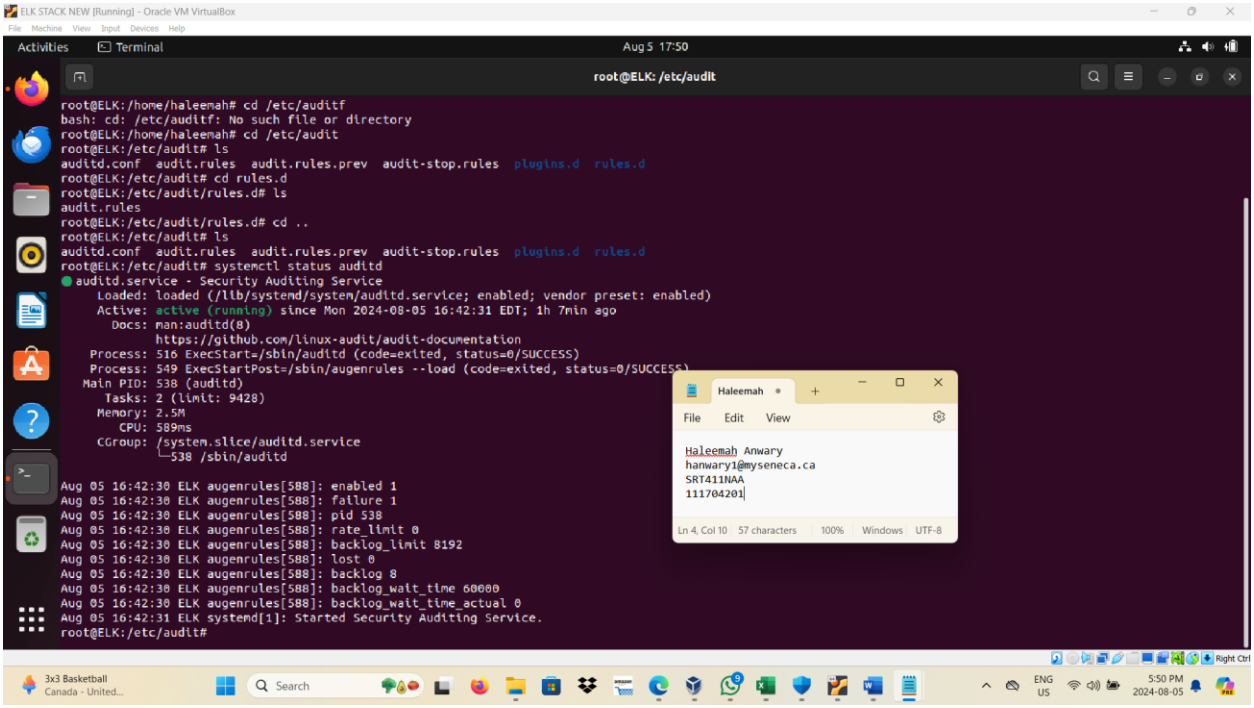
ELK stack VM



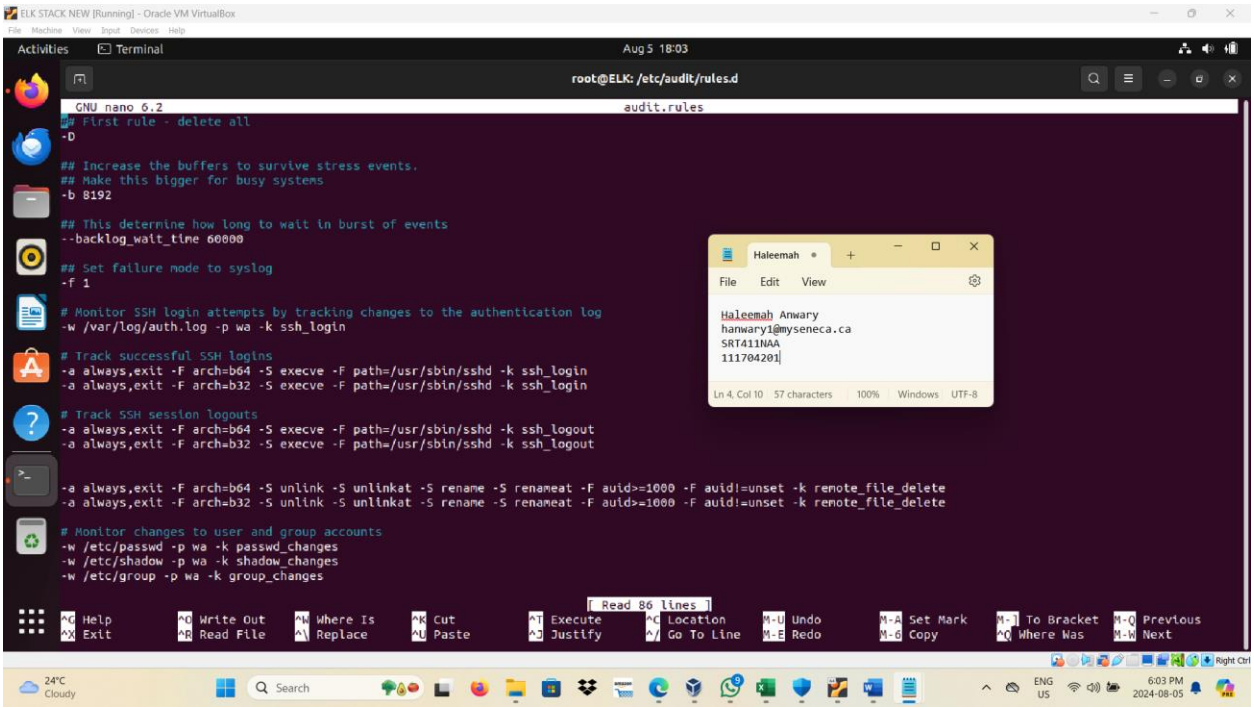
Attacker Kali VM

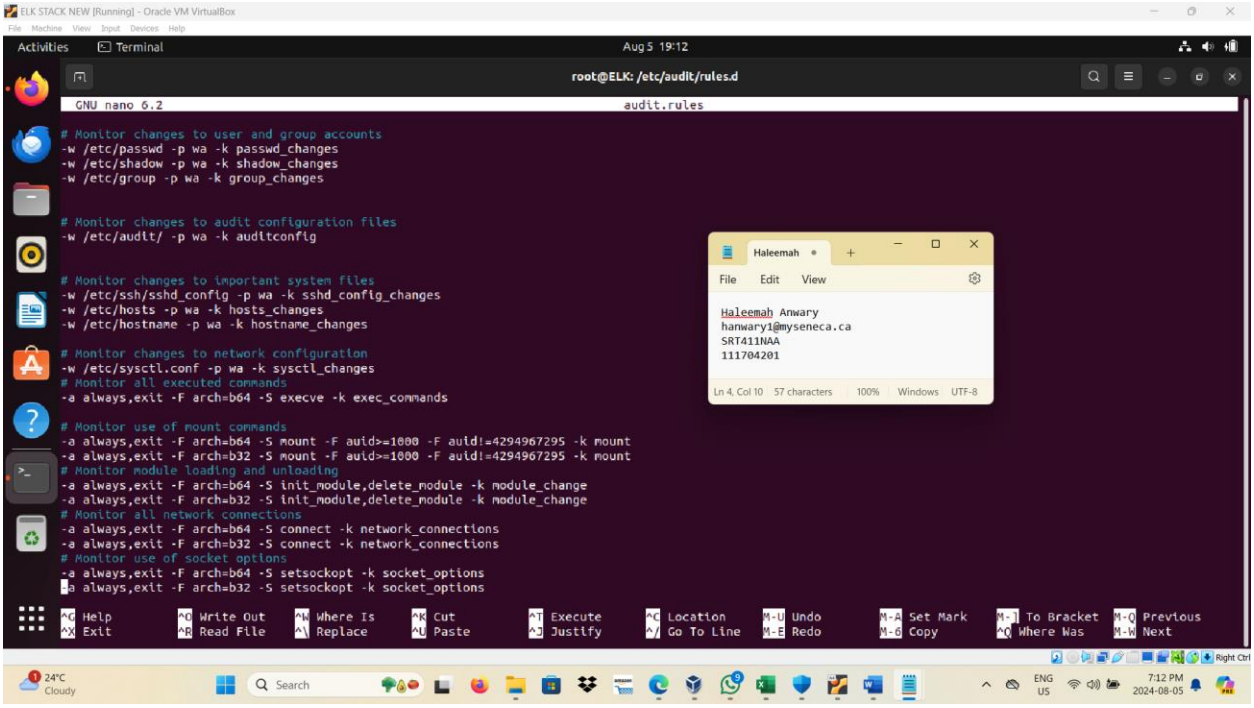


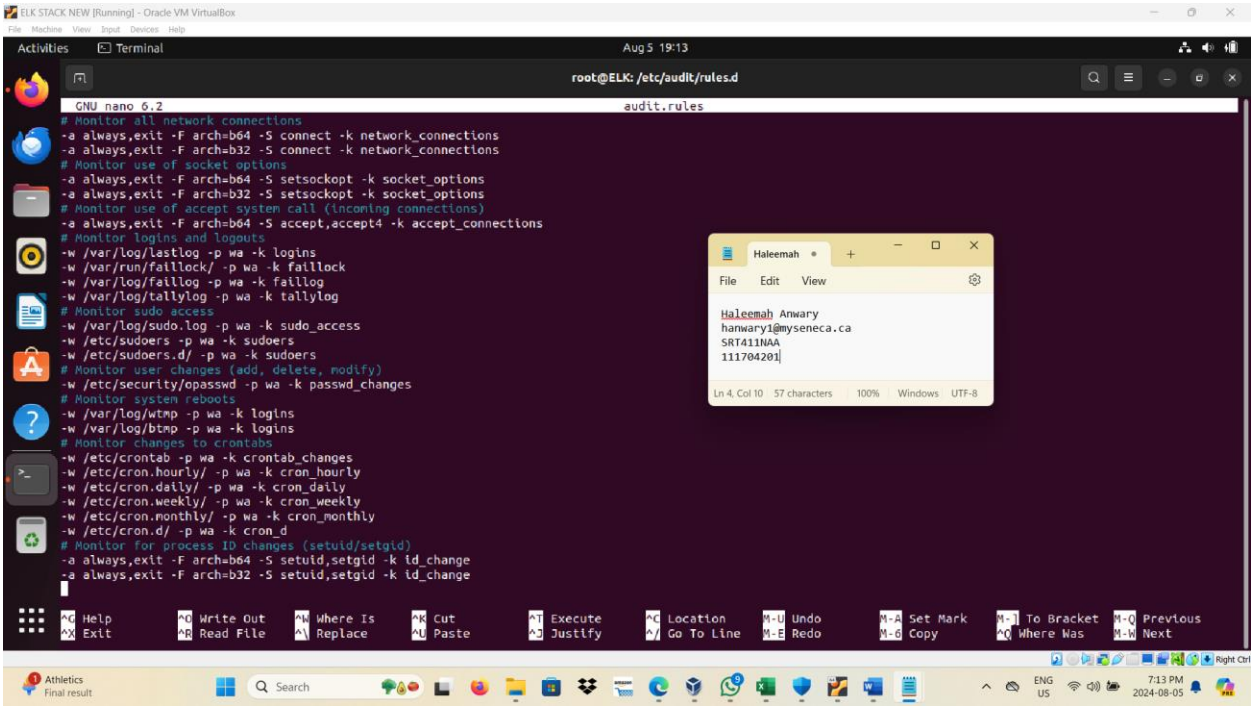
Auditd Installed and Running



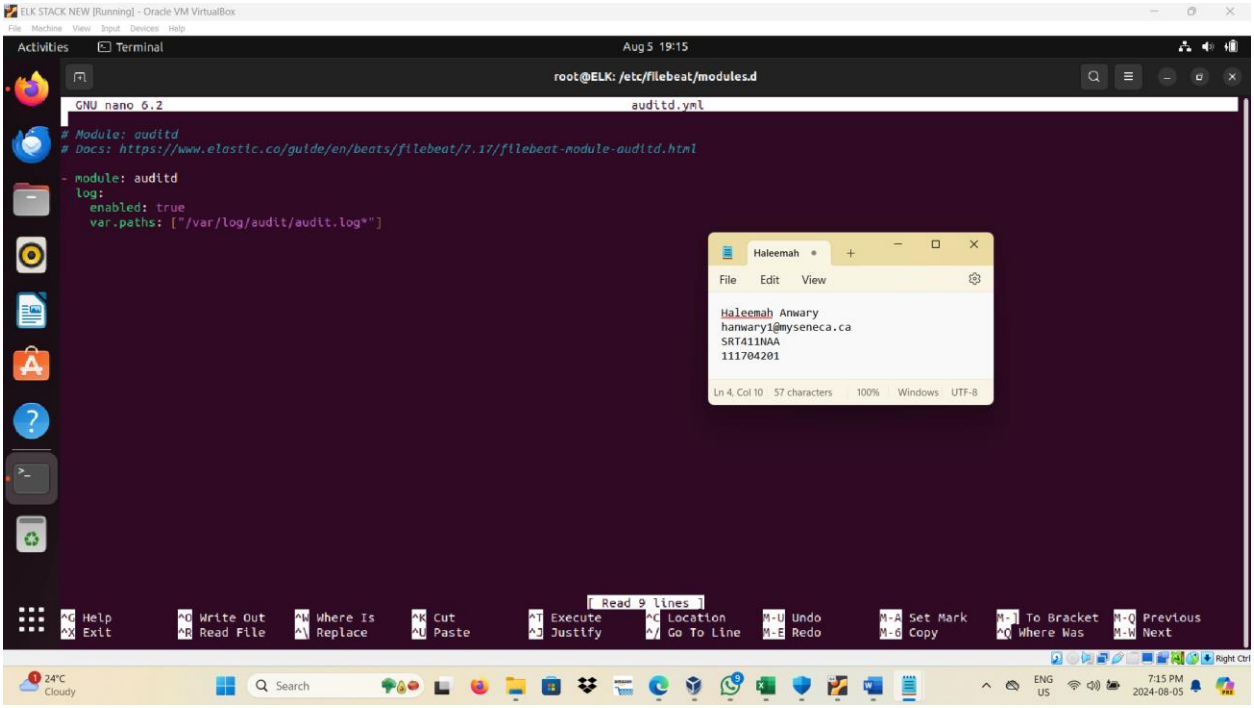
Auditd Rules Configured



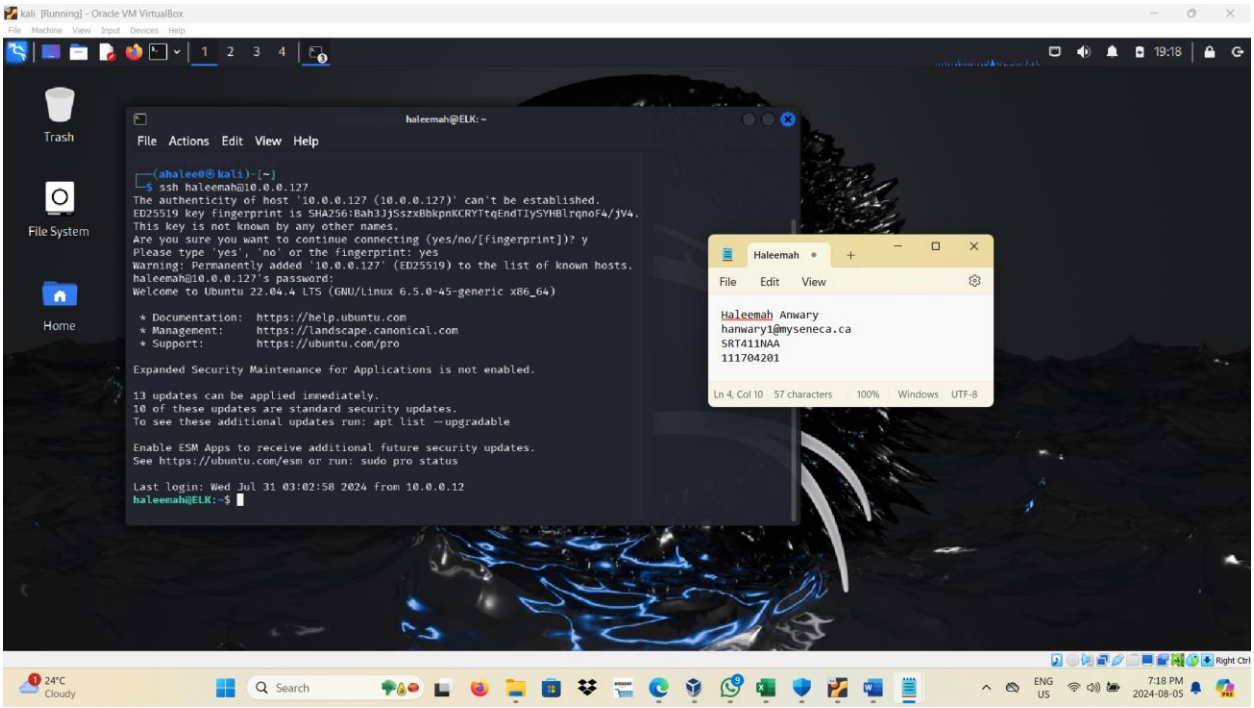

ELK STACK NEW [Running] - Oracle VM VirtualBox
Activities
Terminal
Aug 5 19:12
root@ELK: /etc/audit/rules.d
GNU nano 6.2
audit.rules
Monitor changes to user and group accounts
-w /etc/passwd -p wa -k passwd_changes
-w /etc/shadow -p wa -k shadow_changes
-w /etc/group -p wa -k group_changes
Monitor changes to audit configuration files
-w /etc/audit/ -p wa -k auditconfig
Monitor changes to important system files
-w /etc/ssh/sshd_config -p wa -k sshd_config_changes
-w /etc/hosts -p wa -k hosts_changes
-w /etc/hostname -p wa -k hostname_changes
Monitor changes to network configuration
-w /etc/sysctl.conf -p wa -k sysctl_changes
Monitor all executed commands
-a always,exit -F arch=b64 -S execve -k exec_commands
Monitor use of mount commands
-a always,exit -F arch=b64 -S mount -F auid>=1000 -F auid!=4294967295 -k mount
-a always,exit -F arch=b32 -S mount -F auid>=1000 -F auid!=4294967295 -k mount
Monitor module loading and unloading
-a always,exit -F arch=b64 -S init_module,delete_module -k module_change
-a always,exit -F arch=b32 -S init_module,delete_module -k module_change
Monitor all network connections
-a always,exit -F arch=b64 -S connect -k network_connections
-a always,exit -F arch=b32 -S connect -k network_connections
Monitor use of socket options
-a always,exit -F arch=b64 -S setsockopt -k socket_options
-a always,exit -F arch=b32 -S setsockopt -k socket_options
Haleemah
File Edit View
Haleemah Anwary
hanwary1@myseneca.ca
SRT411NAA
111704201
Ln 4, Col 10 57 characters 100% Windows UTF-8
^G Help ^O Write Out ^W Where Is ^K Cut ^T Execute ^C Location M-U Undo M-A Set Mark M-] To Bracket M-Q Previous
^X Exit ^R Read File ^\ Replace ^U Paste ^J Justify ^/ Go To Line M-E Redo M-6 Copy ^Q Where Was M-W Next
24°C Cloudy
Search
ENG US
7:12 PM 2024-08-05


ELK STACK NEW [Running] - Oracle VM VirtualBox
Activities
Terminal
Aug 5 19:13
root@ELK: /etc/audit/rules.d
GNU nano 6.2
audit.rules
Monitor all network connections
-a always,exit -F arch=b64 -S connect -k network_connections
-a always,exit -F arch=b32 -S connect -k network_connections
Monitor use of socket options
-a always,exit -F arch=b64 -S setsockopt -k socket_options
-a always,exit -F arch=b32 -S setsockopt -k socket_options
Monitor use of accept system call (incoming connections)
-a always,exit -F arch=b64 -S accept,accept4 -k accept_connections
Monitor logins and logouts
-w /var/log/lastlog -p wa -k logins
-w /var/run/faillock/ -p wa -k faillock
-w /var/log/faillog -p wa -k faillog
-w /var/log/tallylog -p wa -k tallylog
Monitor sudo access
-w /var/log/sudo.log -p wa -k sudo_access
-w /etc/sudoers -p wa -k sudoers
-w /etc/sudoers.d/ -p wa -k sudoers
Monitor user changes (add, delete, modify)
-w /etc/security/opasswd -p wa -k passwd_changes
Monitor system reboots
-w /var/log/wtmp -p wa -k logins
-w /var/log/btmp -p wa -k logins
Monitor changes to crontabs
-w /etc/crontab -p wa -k crontab_changes
-w /etc/cron.hourly/ -p wa -k cron_hourly
-w /etc/cron.daily/ -p wa -k cron_daily
-w /etc/cron.weekly/ -p wa -k cron_weekly
-w /etc/cron.monthly/ -p wa -k cron_monthly
-w /etc/cron.d/ -p wa -k cron_d
Monitor for process ID changes (setuid/setgid)
-a always,exit -F arch=b64 -S setuid,setgid -k id_change
-a always,exit -F arch=b32 -S setuid,setgid -k id_change
Haleemah
File Edit View
Haleemah Anwary
hanwary1@myseneca.ca
SRT411NAA
111704201
Ln 4, Col 10 57 characters 100% Windows UTF-8
^G Help ^O Write Out ^W Where Is ^K Cut ^T Execute ^C Location M-U Undo M-A Set Mark M-] To Bracket M-Q Previous
^X Exit ^R Read File ^\ Replace ^U Paste ^J Justify ^/ Go To Line M-E Redo M-6 Copy ^Q Where Was M-W Next
Athletics Final result
Search
ENG US
7:13 PM 2024-08-05

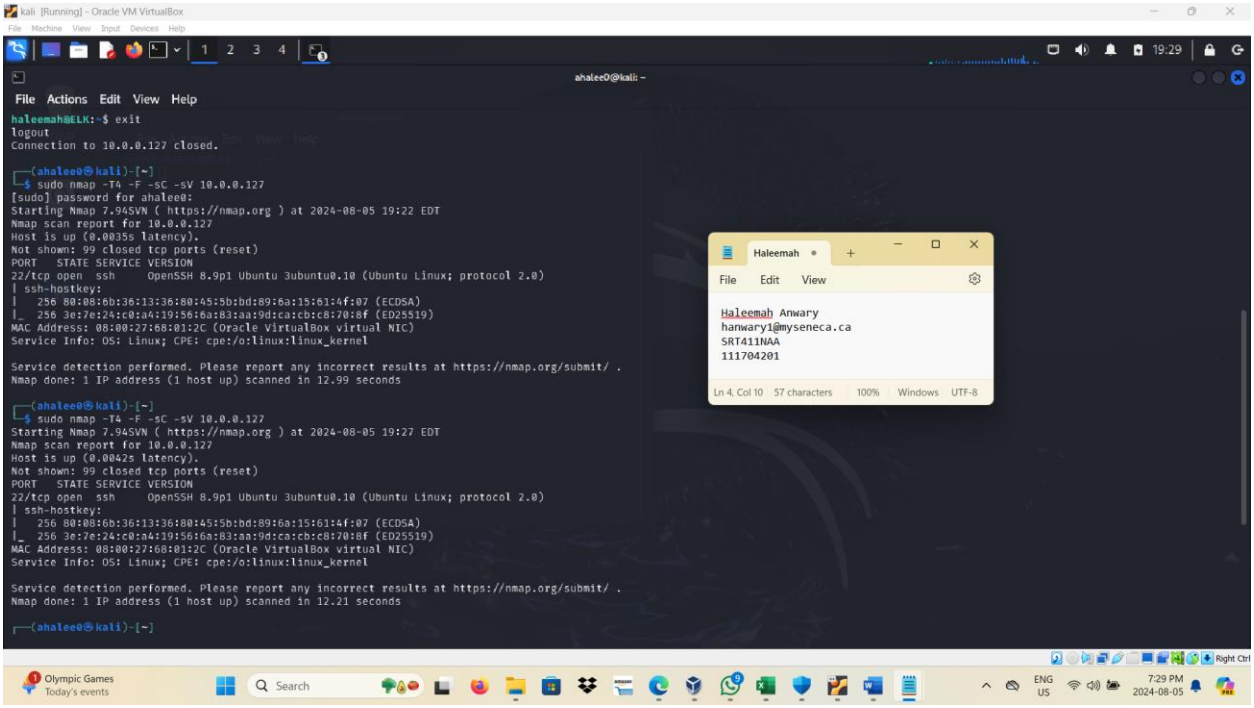
Enabled Filebeat Module and put in the path for the auditd



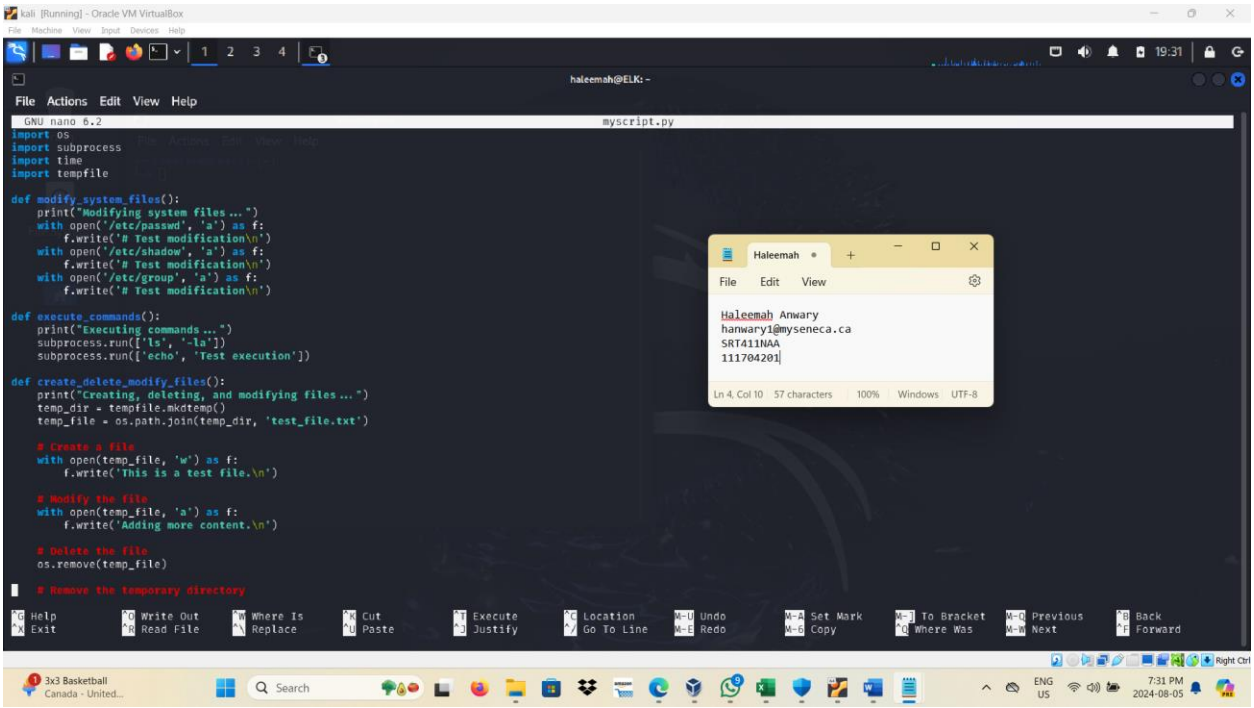
Kali Attack : SSH login



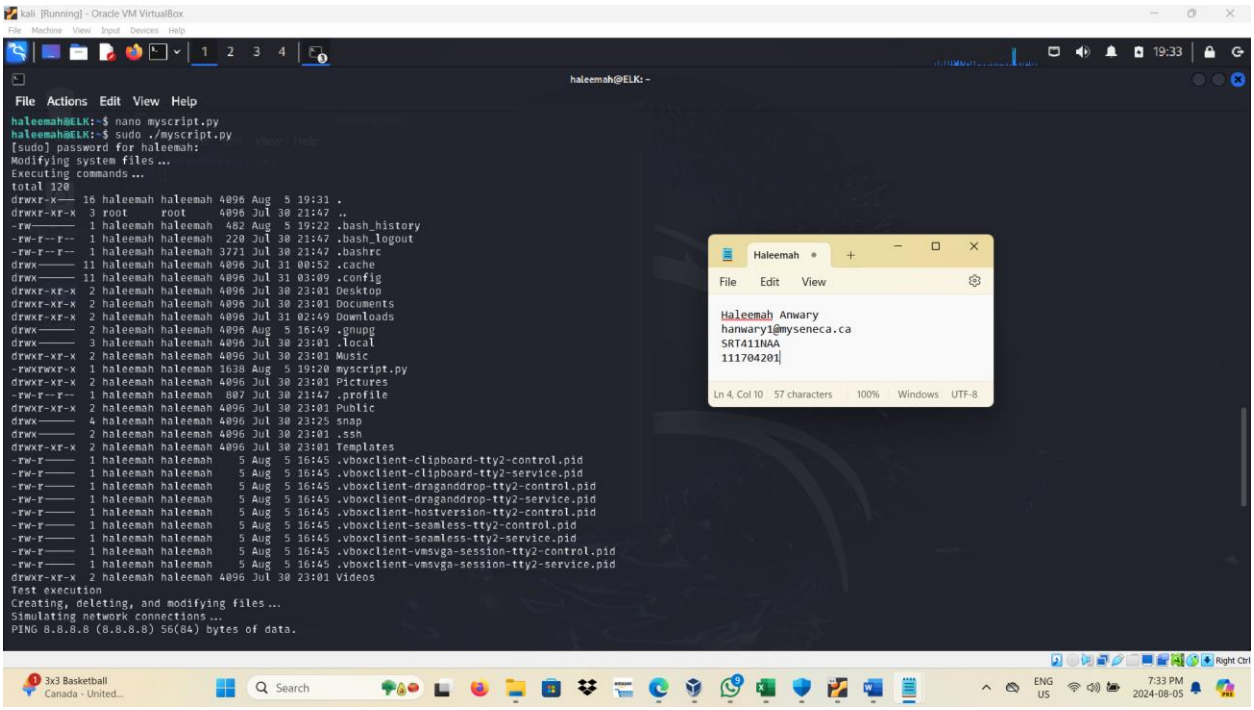
NMAP SCAN



Running script to trigger the system rules

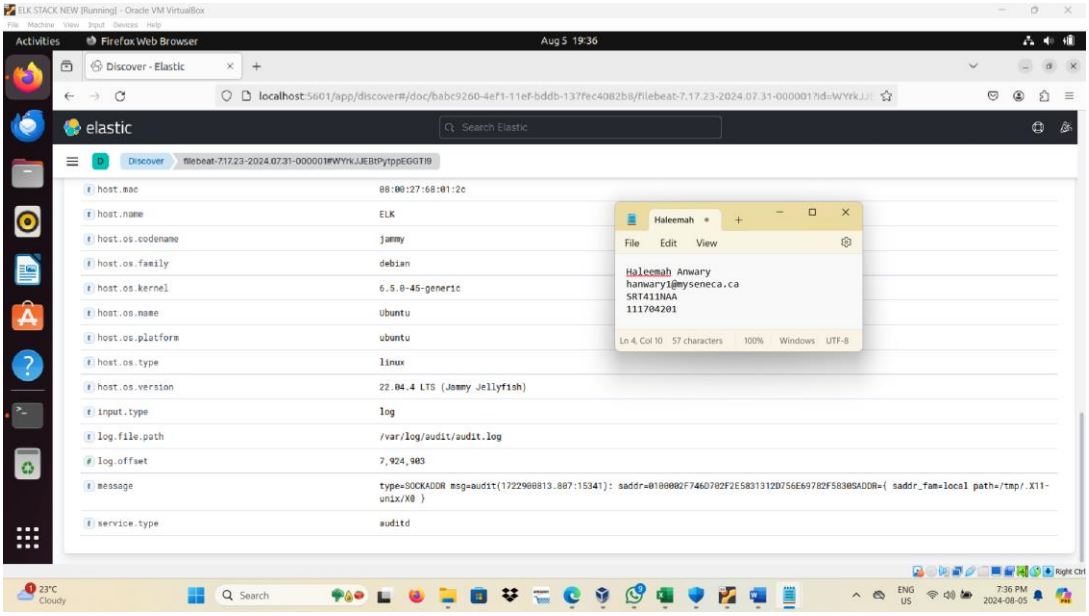


Script Running

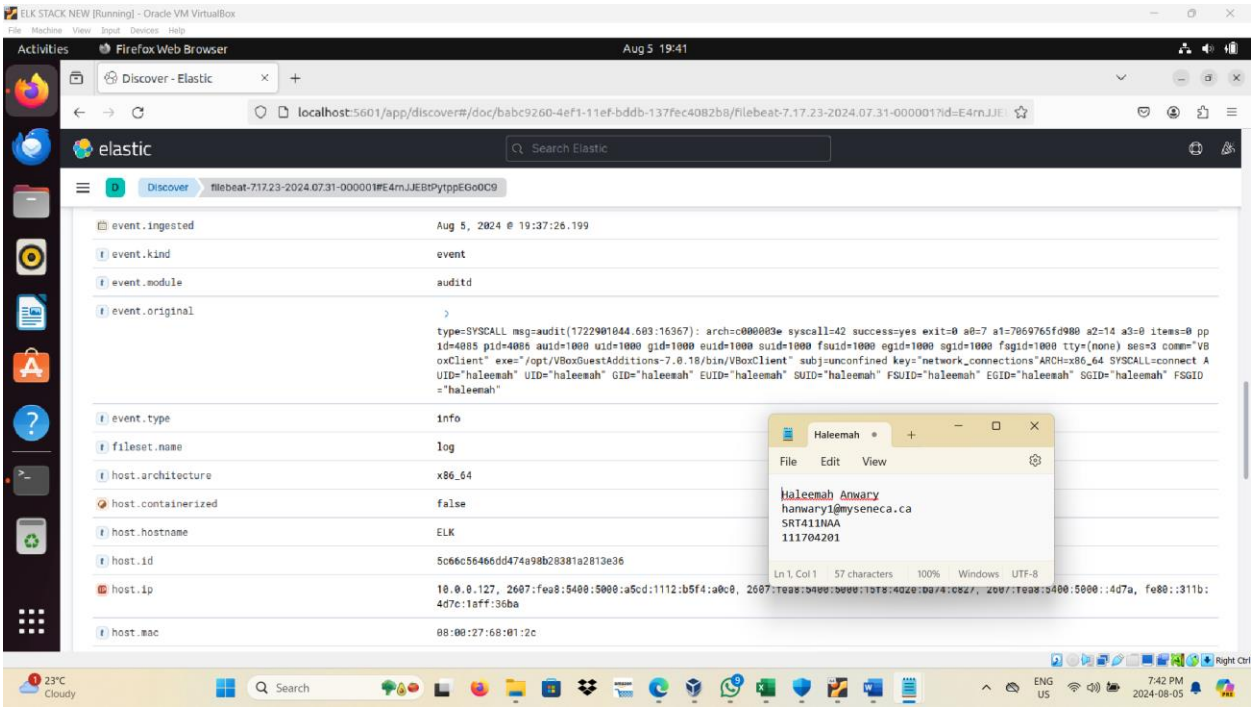


Data in Discover

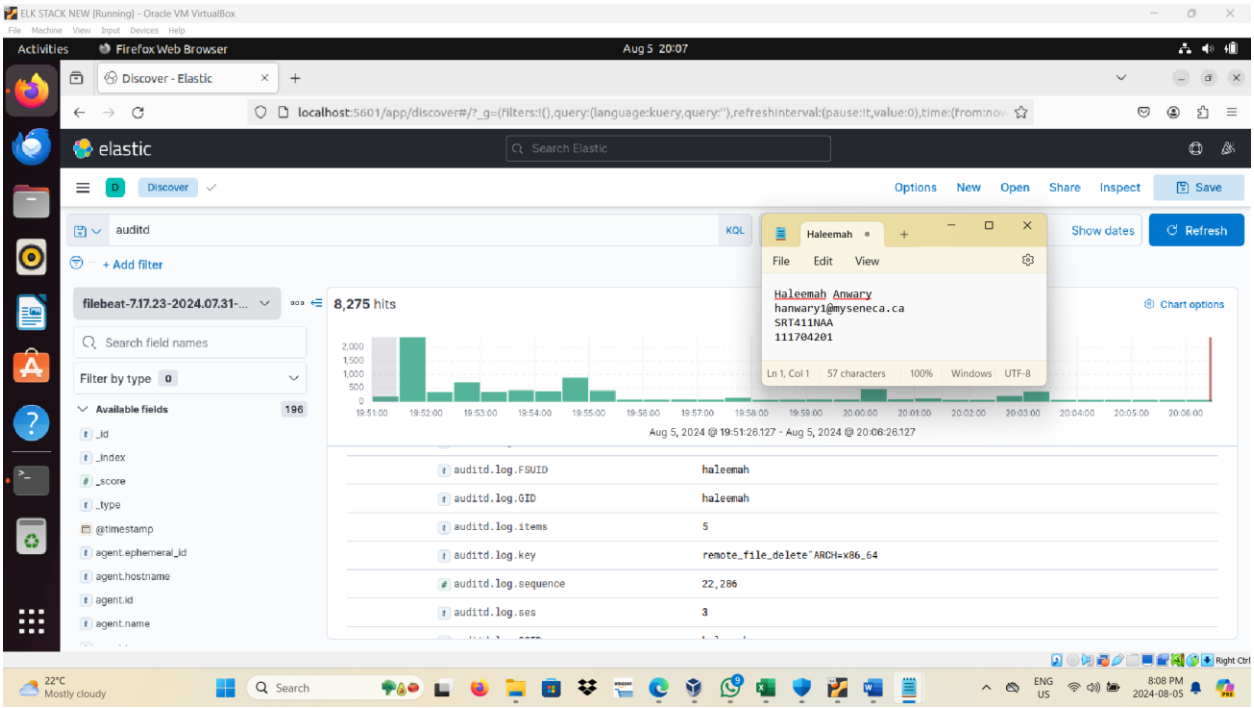
Entries like this can be useful for monitoring and auditing access to the X server, which could be relevant in scenarios involving security or privacy concerns. Unauthorized access or connections to the X server might be indicative of an attempt to capture screen information or interfere with the GUI session.



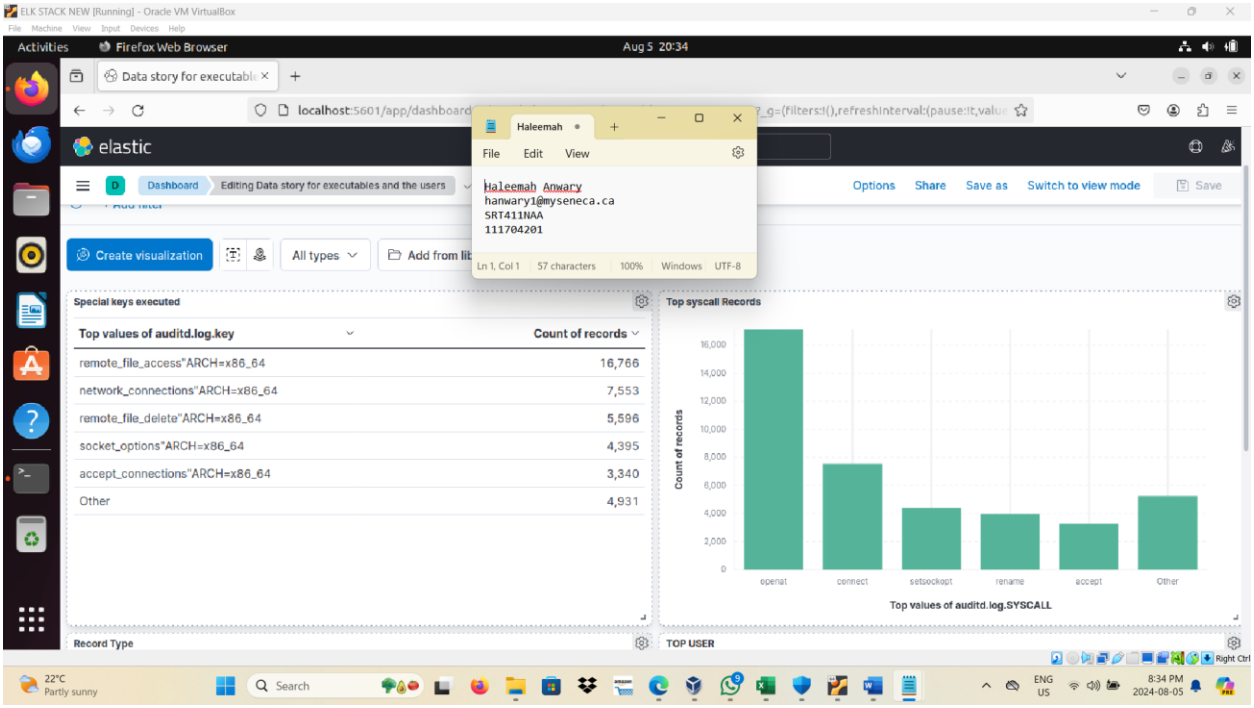
This audit event is recording a network connection attempt by the process VBoxClient, which is a component of the VirtualBox Guest Additions. The process is running as the user "haleemah", and the connection was successfully established (as indicated by success=yes).

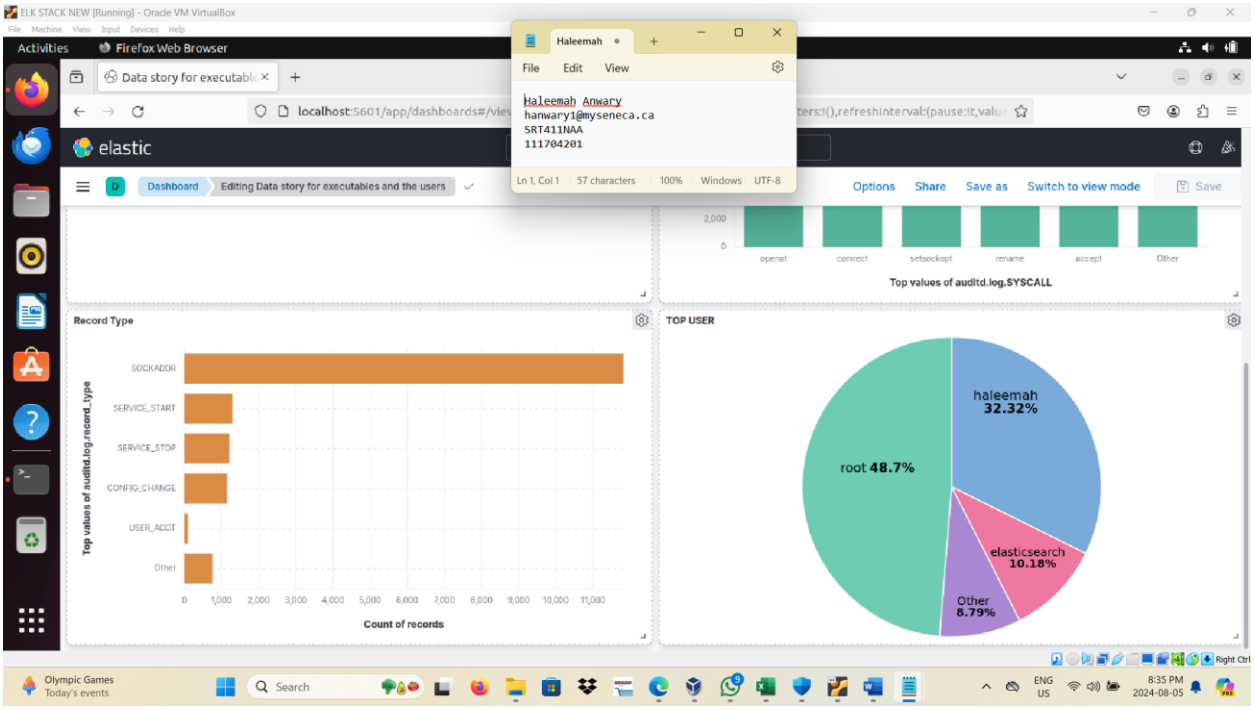


Remote files delete



Dashboard telling the story of executables, keys and the users involved





Task 03 (Questions on Automated Threat Detection and Open-Source Threat Intelligence)

Top Ten Threats in 2022 and Detection Methods

1. Ransomware Attacks
Detection: EDR tools, behavioral analysis, threat intelligence.

2. Phishing Attacks
Detection: Email filtering, user training, anomaly detection.

3. Supply Chain Attacks
Detection: SBOM, code integrity checks, vendor assessments.

4. Zero-Day Exploits
Detection: Threat intelligence, anomaly detection, patch management.

5. Insider Threats
Detection: UEBA, access control monitoring, DLP.

6. Denial of Service (DoS) Attacks
Detection: Network traffic analysis, rate limiting, IPS.

7. Advanced Persistent Threats (APTs)
Detection: Continuous monitoring, threat hunting, behavioral analysis.

8. Credential Stuffing
Detection: MFA, login attempt monitoring, breach detection.

9. Malware
Detection: Antivirus, sandboxing, real-time monitoring.

10. Social Engineering
Detection: User training, simulated phishing, verification processes.

Autonomous Threat Hunting System

An Autonomous Threat Hunting System is a proactive security approach using AI, machine learning, and automation to continuously scan for cyber threats. Unlike traditional methods that rely on human intervention, this system autonomously collects data, identifies patterns, and responds to threats in real-time, helping to protect the network around the clock.

Key Components:

Automated Data Collection: Continuous gathering of network logs and telemetry.

AI/ML Algorithms: Advanced analysis to detect suspicious patterns.

Automated Response: Pre-defined actions like isolating compromised systems or alerting security teams.

Compare Attack-Based vs. Data-Based Threat Hunting

Attack-Based Threat Hunting:

- Targets specific known attack techniques and tactics (TTPs).
- Uses threat intelligence to search for indicators of compromise (IOCs).
- Example: Identifying traces of known malware or suspicious network traffic.

Data-Based Threat Hunting:

- Analyzes large datasets to find anomalies or patterns signaling unknown threats.
- More exploratory, relying on data analytics and machine learning.
- Example: Detecting unusual login times or access patterns that suggest a compromised account.

Visuals and Explanations

These visuals can be explained by describing the types of data being analyzed (e.g., logs, network traffic) and the specific patterns that indicate a threat. Using Kibana's tools to highlight correlations between different events and how they contribute to identifying an ongoing threat.

Classification Rules for Threat Levels

- **Low**: Minor incidents (e.g., failed login attempts).
  Response: Log and monitor.
- **Medium**: More serious but contained incidents (e.g., detected malware).
  Response: Investigate and mitigate.
- **High**: Severe incidents with major impact (e.g., phishing breach).
  Response: Immediate response and mitigation.
- **Critical**: Catastrophic threats (e.g., active ransomware attack).
  Response: Immediate and comprehensive response, possibly involving incident response teams.

Data Collection and Analysis

Data collection in a security context involves gathering information from various sources, including system logs, network traffic and threat intelligence feeds. Analysis involves using this data to identify unusual patterns or indicators of compromise. Techniques such as correlation, anomaly detection, and behavioral analysis are applied to extract meaningful insights from the collected data.

Lessons Learned and Recommendations

Automated threat detection and threat intelligence have proven essential for modern cybersecurity defenses, emphasizing the importance of continuous monitoring and proactive threat hunting to stay ahead of evolving threats. To enhance these defenses, it's recommended to regularly update threat intelligence feeds, invest in training security teams on advanced tools like ELK, and continuously refine incident response plans based on lessons learned from past incidents.

Define Open-Source Threat Intelligence and Its Benefits

Open-Source Threat Intelligence (OSINT) involves gathering and analyzing publicly available information from sources like social media, public databases, forums, and news websites to identify potential threats. The benefits of OSINT include being cost-effective, as many tools and resources are freely available, providing timely information that helps organizations quickly respond to emerging threats, and offering wide coverage by accessing a broad range of data sources for comprehensive threat analysis.

Detailed Explanation of Elastic Agent Integrations and Data Types

Elastic Agent integrates with various data sources to collect logs, metrics, and other relevant data for comprehensive threat detection and analysis. Key integrations include network security, cloud security, and endpoint security tools, which together provide a unified view of security data. The types of data collected by Elastic Agent include logs, which are detailed records of system and application events; metrics, which capture quantitative data such as CPU usage, memory usage, and network latency; and security events, which are alerts and notifications triggered by security systems based on predefined rules.

Visual Explanations from Kibana Dashboards

- **Heatmaps**: To show the intensity of security events across different times or locations.
- **Bar Charts**: To display the frequency of different types of threats detected.
- **Line Graphs**: To illustrate trends over time, such as the rise or fall in the number of detected threats.

These visuals help in quickly understanding the security landscape and identifying areas that need immediate attention.

Comments on Elastic Security for Threat Intelligence

Elastic Security provides a robust platform for integrating and analyzing threat intelligence data, offering centralized management to handle data from multiple sources, real-time analysis tools for detecting threats as they occur, and scalability to manage large volumes of data, making it suitable for organizations of all sizes. Its comprehensive capabilities make Elastic Security a valuable tool for modern cybersecurity operations, providing the essential infrastructure to effectively detect, analyze, and respond to threats.

References

Auditd module | Filebeat Reference [7.17] | Elastic

Threat Intel module | Filebeat Reference [8.14] | Elastic